

Decoding the MoLF Experts: Multi-Modal Semantic Grounding via Pathology VLMs and Biomedical Literature

1. Motivation

The Mixture-of-Latent-Flow (**MoLF**) model [\[link\]](#) establishes a new state-of-the-art in pan-cancer histogenomic prediction by utilizing a **Mixture-of-Experts (MoE)** velocity field. While this architecture effectively decouples diverse tissue patterns, the specific morphological concepts captured by each expert remain **implicit**. Currently, we can observe that experts specialize, but we lack a formal verbal description of their clinical or biological triggers.

This project aims to bridge this gap by leveraging the joint embedding spaces of **Pathology Vision-Language Models (VLMs)** to perform zero-shot literature retrieval. By calculating similarity scores between the visual features of "expert-activating" patches and tokenized biomedical concepts, we can transition from "black-box" routing to interpretable, clinically grounded primitives.

2. Research Objectives

- **Survey** the state-of-the-art in pathology VLMs to identify models with the most robust joint image-text embedding spaces.
- **Quantify** the relationship between morphology and clinical semantics by scoring expert-activating patches against a corpus of tokenized pathology literature.
- **Map** the learned experts to specific biological programs (e.g., Hallmark pathways) through the lens of established medical terminology.

3. Execution Plan

Phase 0: VLM & Literature Corpus Research

- **Task 0.1: VLM Benchmarking.** Research current pathology VLMs. Evaluate them based on their ability to handle joint embeddings for image-text retrieval tasks rather than just simple classification.
- **Task 0.2: Semantic Corpus Assembly.** Identify or compile a "tokenized literature" dataset. This could include: (I don't have enough domain knowledge here. We need refer to other literatures)
 - Standardized vocabularies (e.g., SNOMED-CT, Gene Ontology terms).
 - Pathology textbooks or PubMed abstracts filtered for morphological descriptors (e.g., "pleomorphism," "mitotic activity," "desmoplastic stroma").
- **Task 0.3: Implementation Strategy.** Propose a method for tokenizing this literature into a searchable vector space that is compatible with the selected VLM's text encoder.

Phase I: Activation Mapping & Visual Extraction

- **Task 1.1: Environment Setup.** Utilize the cleaned MoLF codebase(I still need to clean that. Probably the second week of May) to run inference on the HEST-1k pan-cancer benchmark.
- **Task 1.2: Expert Archotyping.** For each of the N=6 experts , identify the "top-K" patches (e.g., top 100) that generate the highest gating logit scores.
- **Task 1.3: Visual Feature Projection.** Extract the morphology features for these patches using the VLM's vision encoder (comparable to the UNI-v2 features used in the original paper).

Phase II: Multi-Modal Similarity Scoring

- **Task 2.1: Semantic Retrieval.** Calculate the cosine similarity (or a similar distance metric) between the mean visual embedding of each expert's top patches and the embeddings of the tokenized literature/concepts.
- **Task 2.2: Statistical Ranking.** Rank the top N biomedical concepts per expert. Analyze whether these rankings remain consistent across different cancer types (e.g., does "Expert 3" always respond to "lymphocytic infiltration" across both Lung and Breast cancer?).
- **Task 2.3: Cross-Modal Alignment.** Compare these textual concepts with the ground-truth gene expression (e.g., Hallmark pathways) predicted by MoLF to see if "visual semantics" and "transcriptomic function" align.

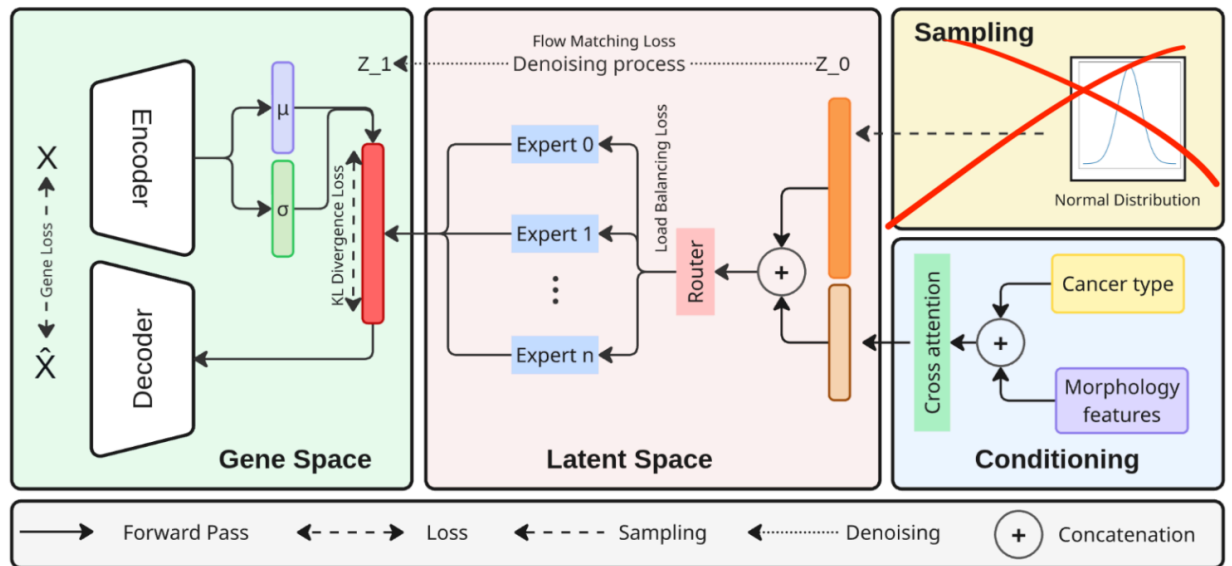
Phase III: Validation & Reporting (Pathologist Review Optional)

- **Task 3.1: Expert Dictionary.** Generate a comprehensive "Clinical Lexicon" for MoLF, naming each expert based on its highest-scoring literature matches.
- **Task 3.2: Error Analysis.** Investigate cases where the VLM similarity scores are low or ambiguous, identifying where current models might struggle with rare morphological variants.

Phase IV: Algorithmic Ablation - Direct Flow Matching

- **Task 4.1: Direct Flow Formulation.** Modify the MoLF architecture to bypass the standard normal distribution prior. Formulate a direct flow matching objective mapping directly from the conditional morphological embeddings (the source) to the target gene expression.
- **Task 4.2: Pipeline Re-Evaluation.** Run the newly trained "direct flow" model through the exact Interpretability Pipeline developed in Phases I and II. Extract the new expert-activating patches and score them against the established biomedical literature corpus.
- **Task 4.3: Comparative Expert Analysis.** Compare the "Clinical Lexicon" generated for the baseline MoLF model with the lexicon of the direct flow model. Analyze how the removal of the noise prior alters the morphological concepts the experts choose to specialize in (e.g., do the experts become more biologically distinct, or do their

specializations collapse?



4. Expected Outcomes

- **A Grounded Model:** A documented map of MoLF's internal logic, translating abstract expert routing into recognized clinical terminology.
- **Interpretability Pipeline:** A reusable framework for grounding any histomorphological feature in medical literature via VLM similarity scoring.
- **Zero-Shot Insights:** Verification of whether the "fundamental biological primitives" the model captures are truly conserved across species and datasets.

5. Future Work & Clinical Translation

If the interpretability pipeline and algorithmic ablation yield compelling, biologically significant findings, we plan to expand the scope of this research toward clinical application.

- **Spatial Transcriptomics Integration:** We will collaborate with domain experts in spatial transcriptomics to rigorously validate the spatial-molecular alignments discovered by the model, ensuring our visual semantics match ground-truth localized gene expression.
- **Internal Dataset Deployment:** To evaluate the model's robustness and clinical utility in a real-world setting, we will deploy the grounded MoLF architecture on our proprietary internal datasets.
- **Medical Journal Submission:** Upon successful validation on internal cohorts and the generation of high-quality visual evidence mapping morphological features to transcriptomic functions, we aim to compile these findings into a comprehensive manuscript for submission to a top-tier medical or bioinformatics journal.

